

Phytochemicals of apple peels: isolation, structure elucidation, and their antiproliferative and antioxidant activities.

Bioactivity-guided fractionation of Red Delicious apple peels was used to determine the chemical identity of bioactive constituents, which showed potent antiproliferative and antioxidant activities. Twenty-nine compounds, including triterpenoids, flavonoids, organic acids and plant sterols, were isolated using gradient solvent fractionation, Diaion HP-20, silica gel, and ODS columns, and preparative HPLC. Their chemical structures were identified using HR-MS and 1D and 2D NMR. Antiproliferative activities of isolated pure compounds against HepG2 human liver cancer cells and MCF-7 human breast cancer cells were evaluated. On the basis of the yields of isolated flavonoids (compounds 18- 23), the major flavonoids in apple peels are quercetin-3-O-beta-D-glucopyranoside (compound 20, 82.6%), then quercetin-3-O-beta-D-galactopyranoside (compound 19, 17.1%), followed by trace amounts of quercetin (compound 18, 0.2%), (-)-catechin (compound 22), (-)-epicatechin (compound 23), and quercetin-3-O-alpha-L-arabinofuranoside (compound 21). Among the compounds isolated, quercetin (18) and quercetin-3-O-beta-D-glucopyranoside (20) showed potent antiproliferative activities against HepG2 and MCF-7 cells, with EC 50 values of 40.9 +/- 1.1 and 49.2 +/- 4.9 microM to HepG2 cells and 137.5 +/- 2.6 and 23.9 +/- 3.9 microM to MCF-7 cells, respectively. Six flavonoids (18-23) and three phenolic compounds (10, 11, and 14) showed potent antioxidant activities. Caffeic acid (10), quercetin (18), and quercetin-3-O-beta-D-arabinofuranoside (21) showed higher antioxidant activity, with EC 50 values of <10 microM. Most tested flavonoids and phenolic compounds had high antioxidant activity when compared to ascorbic acid and might be responsible for the antioxidant activities of apples. These results showed apple peel phytochemicals have potent antioxidant and antiproliferative activities.